Technological University Dublin
Blanchardstown

Academic term	2021/2022
Year of study	
Semester	SEMESTER ONE – January 2022
Date of examination	
Time of examination	

Programme code	**Programme title**	**Module code**
TU653	Higher Certificate in Science in Computing in Information Technology	COMP H2026
TU757	Bachelor of Science in Computing in Information Technology	COMP H2026
TU758	Bachelor of Science in Computing in Digital Forensics & Cyber Security	COMP H2026
TU860	Bachelor of Science (Honours) in Computing	COMP H2026
TU863	Bachelor of Science (Honours) in Computing in Digital Forensics & Cyber Security	COMP H2026

Module title	Information Technology Mathematics

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External Examiner(s)	**Dr. Christopher Meudec** **Dr. Simon Caton**

Instructions to candidates:

1.	**To ensure that you take the correct examination, please check that the module and programme which you are following is listed in the table above.**
2.	**Question 1 is Compulsory. Candidates should attempt ALL parts of Question 1 and any other TWO questions.**
3.	**This paper is worth 100 marks. Question 1 is worth 40 marks. The remaining questions are worth 30 marks each.**
4.	**Remember to mark each number and section clearly and to list the questions which you have attempted in the correct order on the front of your exam script.**
5.	**Show all your work.**

DO NOT TURN OVER THIS PAGE UNTIL YOU ARE TOLD TO DO SO

Question 1: (Compulsory)

Attempt ALL Parts

a) Given the following matrices:

$$\boldsymbol{A} = \begin{bmatrix} \mathbf{1} & \mathbf{3} & \mathbf{2} \\ \mathbf{3} & \mathbf{0} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{1} \end{bmatrix}, \ \boldsymbol{B} = \begin{bmatrix} \mathbf{1} \\ \mathbf{1} \\ \mathbf{2} \end{bmatrix} \text{ and } \boldsymbol{C} = \begin{bmatrix} \mathbf{1} \\ \mathbf{3} \\ \mathbf{1} \end{bmatrix}$$

i. What is the size of each matrix *A* and *B*?

ii. Write down the value of the elements $\boldsymbol{a_{23}}$ and $\boldsymbol{b_{11}}$.

iii. Find the matrix given by $\boldsymbol{A^T(2B + C)}$

(5 marks)

b) Evaluate $\boldsymbol{t(2)}$, $\boldsymbol{t(3)}$ and $\boldsymbol{t(4)}$ for the following recursively defined sequence:

$$\boldsymbol{t(1) = 2}$$

$$\boldsymbol{t(n) = t(n-1) + 3} \qquad \boldsymbol{(n > 1)}$$

(5 marks)

c) Explain what is meant by *sampling*. Include in your answer <u>two</u> methods available to generate a sample from a population.

(5 marks)

d) The rainfall in mm, recorded at 7 weather observation stations in (Leinster) Ireland on a day in November 2021 is: 4.7, 3.7, 4.0, 5.7, 2.7, 1.7, 2.3.

Using this data calculate the

i. mean

ii. standard deviation

(5 marks)

e) Evaluate the following probabilities:

i. If two fair 6-sided dice are thrown, calculate the probability that the SUM of two numbers is greater than 5.

ii. If there are 7 items in a list, how many permutations of 3 are there?

(5 marks)

f) In a group of 5 students represented by A, B, C, D, and E, the following are friends with each other:

A and C

A and D

B and C

C and D

C and E

i. Draw a graph to represent this situation.

ii. Lists the sets V and E for the vertices and edges in G = {V, E}.

iii. Construct an *adjacency matrix* for G.

(5 marks)

g) Calculate the gradient of the curve defined by $\boldsymbol{y = 2x^3 - 6x + 3}$ at the point $(\boldsymbol{3, 1})$.

(5 marks)

h) Evaluate the following indefinite integrals:

i. $\int \left(\boldsymbol{x^{-3}}\right). \boldsymbol{dx}$

ii. $\int \left(\boldsymbol{6x^2 + 4x + 2}\right). \boldsymbol{dx}$

(5 marks)

Total: 40 marks

Question 2: Differentiation & Integration

a) Differentiate the function $y = 2x^2 - 3x + 6$ using first principles. **(6 marks)**

b) Differentiate the following:

i. $y = \frac{5x}{x+2}$ **(4 marks)**

ii. $y = (3x + 1)(4x - 5)$ **(4 marks)**

c) Figure 2.1 shows the graph of $y = x^3 - 12x + 2$.

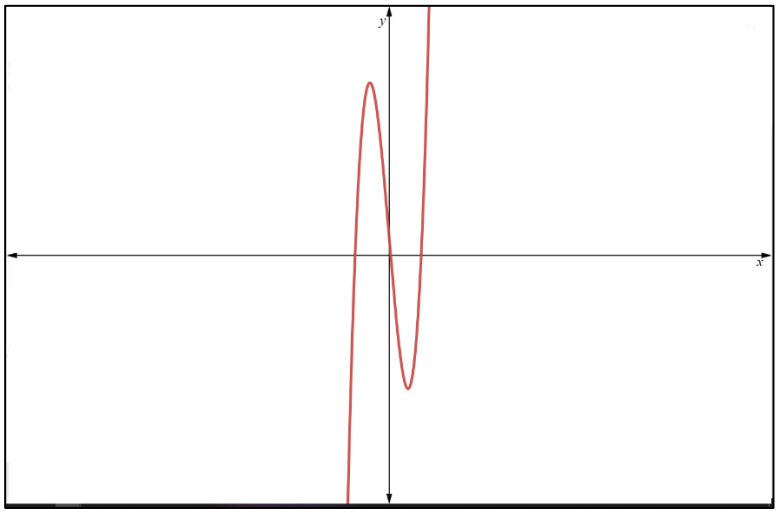


Figure 2.1 Curve given by $y = x^3 - 12x + 2$

i. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ **(4 marks)**

ii. Find the coordinates of the *maximum* and *minimum* turning points of the curve. **(6 marks)**

d) Find the area enclosed by the curve $y = x + 3x^2$ for x between 0 and 2. **(6 marks)**

Total: 30 marks

Question 3: Statistics & Probability

a) Explain how the average of a sample can be obtained. In your answer, discuss the *mean*, *mode* and *median* of a sample and the relationship between these values. **(4 marks)**

b) The box plot in figure 3.1 gives information about the weights of bags on a plane.

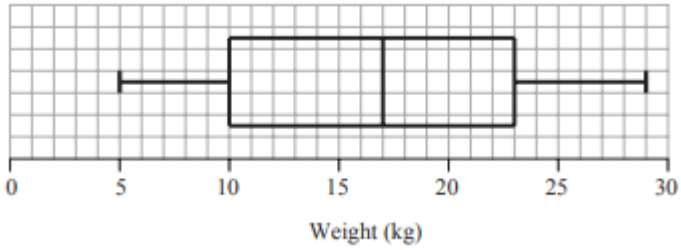


Figure 3.1 box plot of weights of bags on a plane

i. What is the median weight? **(2 marks)**

ii. What is the interquartile range of the weights? **(2 marks)**

c) The following table (table 3.1) represents the distribution of the time taken in minutes m for 40 students to travel to college.

Table 3.1

Time taken (minutes m)	**Number of students**
$0 < m \leq 20$	7
$20 < m \leq 40$	16
$40 < m \leq 60$	11
$60 < m \leq 80$	3
$80 < m \leq 100$	3

i. Calculate the **Mean** of the grouped data. **(3 marks)**

ii. Calculate the **Standard Deviation** of the grouped data. **(5 marks)**

iii. Draw a **Histogram** of the grouped data. **(4 marks)**

iv. Comment on the symmetry or skewness of the distribution of the histogram in part iii. **(2 marks)**

Question continued over page.

d) The Probability distribution for a random variable X is given in table 3.2 below. What is the missing probability?

X	5	10	15	20	25	30
$P[X]$	0.02	0.21	0.17	0.46	0.04	?

(2 marks)

e) A group of students sat a Mathematics exam. Their marks in the exam are normally distributed with a mean of 65 and a standard deviation of 9. If the pass mark for the exam is 40, what is the probability that a student chosen at random has passed the exam. **(6 marks)**

Total: 30 marks

Question 4: Matrices, Graphs & Trees

a) Given the following matrices:

$$\boldsymbol{A} = \begin{bmatrix} \mathbf{5} & \mathbf{2} \\ \mathbf{2} & \mathbf{1} \end{bmatrix} \text{ and } \boldsymbol{B} = \begin{bmatrix} \mathbf{9} & \mathbf{12} \\ \mathbf{4} & \mathbf{5} \end{bmatrix}$$

Find the matrix $\boldsymbol{X}$ that satisfies the equation $\boldsymbol{AX} = \boldsymbol{B}$ **(7 marks)**

b) Let $\boldsymbol{A} = \begin{bmatrix} \mathbf{1} & \mathbf{3} & \mathbf{2} \\ \mathbf{3} & \mathbf{0} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{1} \end{bmatrix}$

i. Write down the *Identity* matrix for $\boldsymbol{A}$ **(2 marks)**

ii. Find the *determinant* of $\boldsymbol{A}$ **(4 marks)**

c) Describe the difference between *breadth-first* and *depth-first* graph traversals. Consider the following graph *G* (figure 4.1), starting at node *D*, in what order will the nodes be visited using breadth-first search? Assume, that if there is a choice between nodes, the algorithm will choose alphabetically.

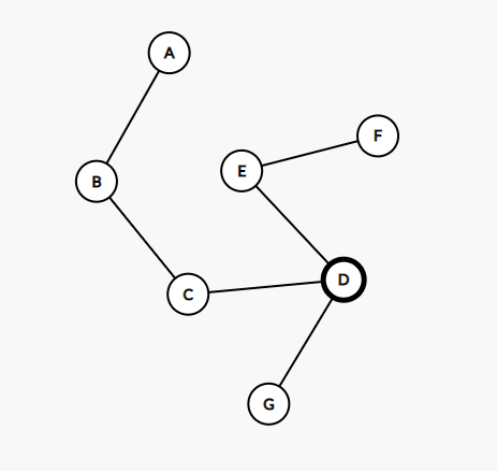


Figure 4.1 Graph G

(9 marks)

Question continued over page.

d) Using Prim's Algorithm, starting at vertex *A*, find the *Minimal Spanning Tree* (MST) for the weighted graph G in figure 4.2 below.

(8 marks)

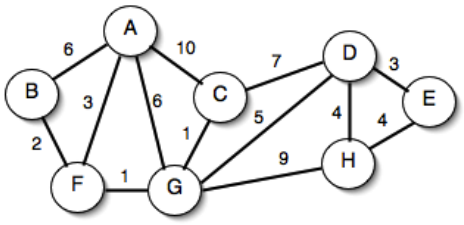


Figure 4.2 Weighted Graph G

Total: 30 marks

Formulae

Matrix Determinant

$\det A = ad - bc$ (2x2 matrix)

$\det A = a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$ (3x3 matrix using row 1)

Matrix Inverse

$A^{-1} = \frac{1}{\det A}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (2x2 matrix)

$A^{-1} = \frac{1}{\det A}C^T$, where *C* is the matrix of Cofactors (3x3 matrix)

General Probability:

If two events A and B are mutually exclusive, then the following law holds:

$P[A \text{ or } B] = P[A] + P[B]$

If the two events are not mutually exclusive, the law above is broadened so that the probability of A or B occurring is given by:

$P[A \text{ or } B] = P[A] + P[B] - P[A \text{ and } B]$

Consider two events *A* and *B* with event *B* dependent on *A*. The probability of the occurrence of both events is given by:

$P[A \text{ and } B] = P[A] \times P[B \mid A]$

Probability Distributions:

$P[X = r] = {}^nC_r\, p^r q^{n-r} = \frac{n!}{(n-r)!r!} p^r q^{n-r}$ (Binomial distribution)

$P[X = r] = \frac{e^{-\mu}\mu^r}{r!}$, where $\mu = np$ (Poisson distribution)

$a = \frac{x - \mu}{\sigma}$ (Normal distribution)

Statistics:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$ (Dataset mean)

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}} \quad \text{or} \quad s^2 = \frac{\sum_{i=1}^{n} x_i^{\,2} - n\left(\bar{x}\right)^2}{n-1}$$ (Dataset standard deviation)

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i m_i}{\sum_{i=1}^{n} f_i}$$ (Grouped data mean)

$$s = \sqrt{\frac{\sum_{i=1}^{n} f_i (m_i - \bar{x})^2}{\sum_{i=1}^{n} f_i - 1}} \quad \text{or} \quad s^2 = \frac{\sum_{i=1}^{M} f_i m_i^{\,2} - M\left(\bar{x}\right)^2}{M-1}$$ (Grouped standard deviation)

Derivation:

First Principles: $\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ **Power Rule:** $\frac{dy}{dx} = nx^{n-1}$

Product Rule: $\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$ **Quotient Rule:** $\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

Integral: $\int \boldsymbol{ax^n dx} = \boldsymbol{\frac{ax^{n+1}}{n+1} + c}, \ \boldsymbol{n \neq -1}$

Definite Integral:

$$\int_a^b f'(x)dx = f(x)\Big]_a^b = f(b) - f(a)$$

Table of Probabilities – P[*Z* > a], where *Z* is the standard normal variable.

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000